“Intelligent Water Distribution Monitoring and Leakage Detection Platform”

CO1 Assessment Tool 2

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DECLARATION

I, **Lenin K 192321050,** of the **Department of Information Technology**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **"Collaborative Story/ Screenplay Writing Platform with Version History"** is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place:

Chennai

Date:

Signature of the Students

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1. Introduction

The Intelligent Water Distribution Monitoring and Leakage Detection Platform is a cloud-based, IoT-enabled software system designed to give water utility operators continuous, real-time visibility into the condition of their distribution networks. The system ingests live telemetry from flow, pressure, and acoustic sensors deployed across the pipeline network, automatically detects and localizes probable leaks through anomaly-detection analytics, and predicts high-risk pipe segments before they fail using historical data and machine learning.

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements, actors, use cases, data requirements, external interfaces, system features, assumptions, and constraints of the proposed platform, following a scenario-based requirement analysis approach.

2. Problem Description

Water utilities typically rely on periodic manual meter readings, physical pipeline inspections, and customer complaints to identify leaks in the distribution network. These traditional methods create serious operational challenges:

- Leaks often go undetected for days or weeks, resulting in significant non-revenue water (NRW) loss and wasted treatment costs.
- There is no continuous, real-time visibility into pipeline pressure and flow conditions across the network.
- Manual inspection is labor-intensive, slow, and inherently reactive rather than proactive.
- Utilities struggle to prioritize which aging or high-risk pipe segments need preventive maintenance or replacement first.
- Field crews are frequently dispatched without precise leak-location data, forcing large-area excavation to locate the actual fault.
- Sensor data, GIS pipeline maps, and maintenance/repair records are managed in disconnected systems with no single source of truth.

The proposed system addresses these issues by providing a single operational workspace with continuous sensor monitoring, automated leak detection and localization, predictive risk scoring, GIS-based visualization, and integrated alerting and work-order management for every stakeholder in the leak-response lifecycle.

3. Scope of the System

The system scope covers the complete monitoring-to-repair lifecycle for a water distribution network: utility and District Metered Area (DMA) setup, IoT sensor registration and real-time data ingestion, live dashboard monitoring, automated leak-anomaly detection and localization, predictive pipe-failure risk scoring, GIS-based network visualization, instant alerting, work-order creation and field-crew management, and compliance/management reporting.

The system is a web-based application with a companion mobile-friendly interface for field technicians. It does not cover water-treatment plant process control (e.g., chemical dosing, filtration control), customer billing or consumer-side metering, or the manufacturing/physical installation of IoT sensor hardware, which lie outside the boundary of this platform.

4. Stakeholders, Business Domain and System Objectives

4.1 Business Domain

The system operates in the water-utility infrastructure and smart-city IoT domain, supporting municipal water boards, regional utility companies, and infrastructure asset-management teams responsible for treated-water distribution networks.

4.2 Stakeholders

- Control Room Operators — monitor live sensor data, validate leak alerts, and dispatch field crews.
- Field Technicians — receive and act on work orders to locate and repair leaks.
- Utility / Asset Managers — review risk scores, prioritize maintenance budgets, and track NRW performance.
- GIS Engineers — maintain the digital twin of the pipeline network and keep spatial data synchronized.
- Data Scientists / ML Engineers — build and tune the leak-detection and pipe-failure-risk models.
- Regulatory Authorities — receive periodic water-loss and compliance reports from the utility.
- Platform Administrators — manage utility accounts, user roles, zone configuration, and system security.
- Product Owner / Organization — owns the platform and its business goals.

4.3 System Objectives

- Provide continuous, real-time visibility into flow, pressure, and acoustic conditions across the distribution network.
- Automatically detect and localize probable leaks with minimal manual monitoring effort.
- Predict high-risk pipeline segments before failure to support proactive maintenance planning.
- Convert detected leaks into actionable, trackable work orders for field repair crews.
- Provide GIS-based visualization of the pipeline network integrated with live sensor and leak data.

- Support regulatory and management reporting on water loss, leak-response times, and asset condition.

5. Functional Requirements

ID	Functional Requirement	Description
FR-01	User Registration & Login	The system shall allow users to register, log in, and manage their profile using secure authentication (including optional multi-factor authentication).
FR-02	Utility / Zone (DMA) Setup	The system shall allow a Utility Admin to define District Metered Areas (DMAs) and pipeline zones and associate sensors with each zone.
FR-03	Sensor Registration & Data Ingestion	The system shall allow registration of IoT flow, pressure, and acoustic sensors and shall continuously ingest their streaming telemetry.
FR-04	Real-Time Monitoring Dashboard	The system shall display a live dashboard of flow and pressure readings across all DMAs, refreshed continuously.
FR-05	Sensor Health Monitoring	The system shall automatically flag sensors that go offline or report faulty/out-of-range readings.
FR-06	Automatic Leak Anomaly Detection	The system shall continuously analyze flow and pressure data for statistical anomalies indicative of a possible leak.
FR-07	Leak Localization & Confidence Scoring	The system shall highlight the probable leak location on the network map with a calculated confidence score.
FR-08	Alert Validation	The system shall allow an operator to review the sensor trend behind a leak alert and mark it as confirmed leak, false positive, or scheduled maintenance.
FR-09	Pipe Failure Risk Scoring	The system shall calculate a failure-risk score for each pipeline segment based on age, material, pressure history, and past incidents.
FR-10	Risk Ranking View	The system shall display a ranked list of high-risk pipeline segments to support maintenance planning.
FR-11	Instant Alerting	The system shall notify operators via push notification, SMS, and email within seconds of a high-confidence leak detection.
FR-12	Work Order Creation & Assignment	The system shall allow an operator to create a work order from a leak alert and assign it to a field technician.
FR-13	Mobile Work-Order Management	The system shall allow field technicians to view assigned work orders and update status (En Route, In Progress, Resolved) with notes and photos on a mobile-friendly interface.
FR-14	GIS Network Visualization	The system shall display the full pipeline network on a GIS map with pipe age, material, and diameter.
FR-15	Pipeline Geometry Import	The system shall allow import and update of pipeline network geometry from shapefiles/GeoJSON.
FR-16	Reporting & Export	The system shall allow generation and export of water-loss and leak-response reports in PDF and CSV formats.
FR-17	Role-Based Access Control	The system shall support roles (Admin, Operator, Field Technician, Manager, Guest) with distinct permissions.
FR-18	Administration Console	The system shall allow Administrators to manage user accounts, zone configuration, and system-wide settings.

6. Non-Functional Requirements

Category	Requirement
Performance	Sensor readings shall be ingested and reflected on the live dashboard within 5 seconds under normal network conditions; leak alerts shall be raised within 30 seconds of a confirmed anomaly.
Scalability	The system shall support at least 50,000 concurrently connected sensors and horizontal scaling of ingestion and processing services during peak load.
Security	All data in transit shall be encrypted using TLS 1.2+ and data at rest shall be encrypted using AES-256; access shall be governed by role-based permissions.
Reliability	The system shall guarantee 99.9% uptime and shall not lose committed sensor or leak-event data even in the event of a server failure (durable storage/backups).
Usability	The monitoring dashboard and mobile work-order interface shall be usable by non-technical operators and field technicians with minimal training.
Maintainability	The system shall follow a modular, service-oriented architecture to allow independent updates to the ingestion, detection, GIS, and notification services.
Availability	Core monitoring, alerting, and leak-detection features shall remain available 24/7, since leaks can occur at any time, with scheduled maintenance windows communicated in advance.
Portability	The web client shall function consistently across major modern browsers (Chrome, Firefox, Edge, Safari) and the mobile interface shall support responsive layouts for field use.
Auditability	Every leak-confirmation action, work-order change, and access-control modification shall be logged with user ID, timestamp, and action performed.

7. Actors and Use Cases

7.1 Primary and Secondary Actors

Actor	Type	Description
Control Room Operator	Primary	Monitors live sensor data, validates leak alerts, and dispatches field crews.
Field Technician	Primary	Receives assigned work orders; locates, repairs, and reports on leaks in the field.
Utility / Asset Manager	Primary	Reviews risk scores, prioritizes maintenance, and tracks NRW performance and reports.
GIS Engineer	Primary	Maintains and imports pipeline network geometry and spatial data.
Platform Administrator	Primary	Manages user accounts, zone configuration, and system settings.
Notification Service	Secondary (System Actor)	Automated component that dispatches push, SMS, and email notifications.
Regulatory Authority / Guest	Secondary	Views shared reports or dashboards in read-only mode; cannot edit.

7.2 Use Case Diagram

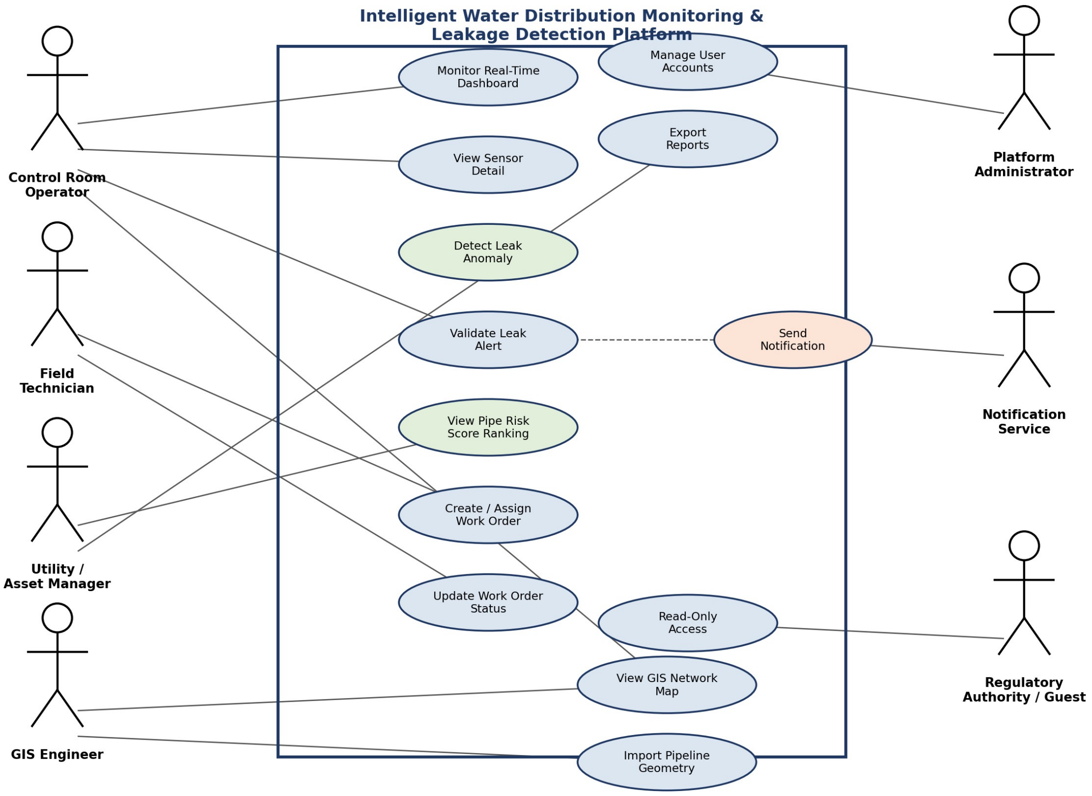

Fig. 1 — Use Case Diagram: Intelligent Water Distribution Monitoring and Leakage Detection Platform

7.3 Use Case Descriptions

Use Case	Primary Actor	Description
Monitor Real-Time Dashboard	Control Room Operator	Actor views live flow/pressure status across all DMAs, color-coded by anomaly status.
View Sensor Detail	Control Room Operator	Actor drills down from a zone into an individual sensor's historical readings.
Detect Leak Anomaly	System (automated)	System continuously analyzes sensor streams and raises a flagged anomaly with a confidence score.
Validate Leak Alert	Control Room Operator	Actor reviews the anomaly trend and marks it as confirmed leak, false positive, or maintenance.
View Pipe Risk Score	Utility / Asset Manager	Actor views pipeline segments ranked by predicted failure risk.
Create / Assign Work Order	Control Room Operator	Actor creates a work order from a confirmed leak alert and assigns it to a technician.
Update Work Order Status	Field Technician	Actor updates job status and attaches repair notes/photos from the field.
View GIS Network Map	Control Room Operator / GIS Engineer	Actor views the pipeline network and active leak locations on a geospatial map.
Import Pipeline Geometry	GIS Engineer	Actor imports or updates pipeline geometry from shapefiles/ GeoJSON.
Export Reports	Utility / Asset Manager	Actor generates and downloads water-loss and leak-response reports.
Send Notification	Notification Service	System dispatches push/SMS/email alerts triggered by leak detection or work-order events.
Manage User Accounts	Platform Administrator	Actor creates, disables, or configures roles for user accounts.
Read-Only Access	Regulatory Authority / Guest	Actor views shared dashboards or reports without edit permissions.

8. Data Requirements

- User Data: user ID, name, email, hashed password/auth token, role, profile details.
- Utility / Zone Data: utility ID, DMA/zone ID, boundary geometry, associated sensor list.
- Sensor Data: sensor ID, type (flow/pressure/acoustic), zone tag, reading value, timestamp, status (online/offline/faulty).
- Leak / Anomaly Records: anomaly ID, sensor/zone reference, detection timestamp, confidence score, validation status (confirmed/false positive/maintenance).
- Risk Score Data: pipe segment ID, age, material, pressure history summary, incident history, calculated risk score.
- Work Order Data: work order ID, linked leak/anomaly ID, assigned technician, status, notes, photos, timestamps.
- GIS / Pipeline Geometry Data: segment ID, coordinates, diameter, material, installation date.
- Notification Data: event type, recipient, delivery channel, read/unread status.
- Audit Log Data: action type, actor, target object, timestamp, IP/session metadata.

9. External Interface Requirements

9.1 User Interfaces

- Responsive web application with a real-time monitoring dashboard, GIS map viewer, alert/work-order panels, and reporting console; a mobile-friendly interface for field technicians.

9.2 Hardware Interfaces

- IoT flow, pressure, and acoustic sensors with network/cellular connectivity; standard client devices (desktop, laptop, tablet, smartphone) with a modern web browser.

9.3 Software Interfaces

- Authentication provider (OAuth 2.0 / SSO integration for organizational login).
- SMS/Email gateway for alert notifications.
- GIS/mapping service for rendering pipeline geometry and sensor locations.
- Cloud object storage service for exported reports and attachments.
- Machine-learning model-serving service for anomaly detection and risk scoring.

9.4 Communication Interfaces

- HTTPS/REST APIs for standard client-server operations.
- MQTT-based protocol for high-frequency IoT sensor telemetry ingestion.
- WebSocket-based protocol for real-time dashboard updates.

10. System Features

Feature	Priority	Summary
Real-Time Sensor Monitoring	High	Continuous ingestion and live dashboard display of flow, pressure, and acoustic data across all zones.
Leak Detection & Localization	High	Automated anomaly detection with confidence-scored leak location highlighted on the map.
Alerting & Work-Order Management	High	Instant multi-channel alerts and end-to-end work-order tracking from detection to repair.
Predictive Maintenance	High	Machine-learning risk scoring of pipeline segments to prioritize preventive maintenance.
GIS Mapping	Medium	Geospatial visualization of the pipeline network integrated with sensor and leak data.
Reporting & Export	Medium	One-click generation of water-loss and leak-response reports in PDF/CSV.
Sensor Health Monitoring	Medium	Automatic flagging of offline or faulty sensors to keep monitoring data trustworthy.
Admin Console	Low	Account management, zone configuration, and system-wide settings for administrators.

11. System Architecture (Supporting Diagram)

The following layered architecture diagram illustrates how the client, application, data, and infrastructure layers interact to deliver real-time sensor monitoring, leak detection, and predictive maintenance.

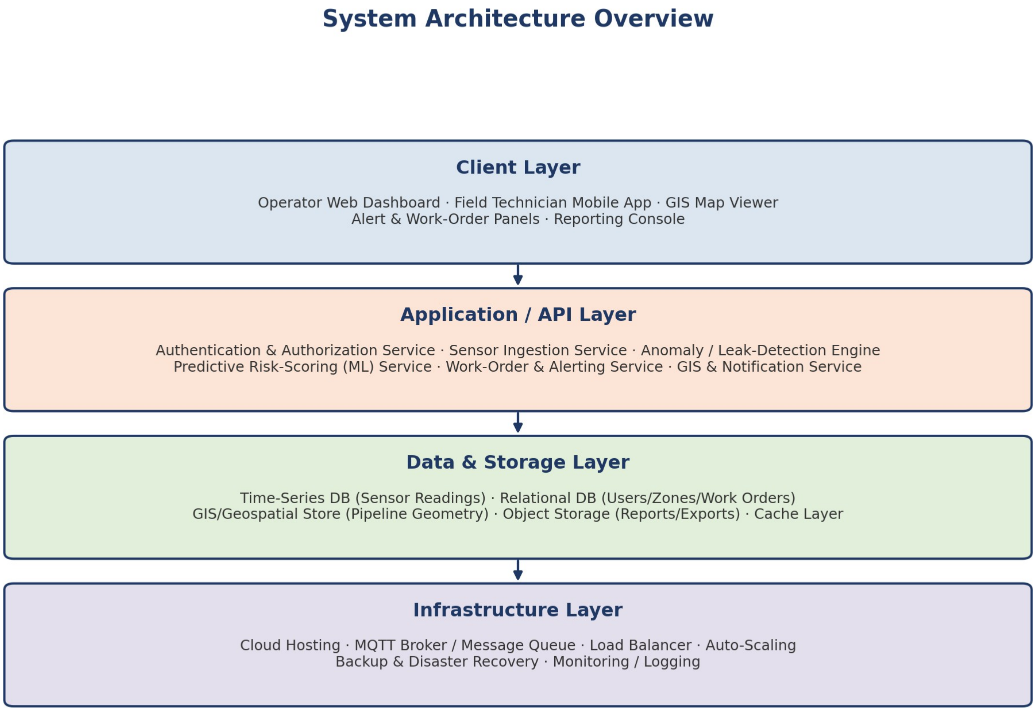

Fig. 2 — High-Level Layered System Architecture

12. Assumptions

- IoT sensors are properly calibrated and physically maintained by the utility or its hardware vendor.
- Each DMA has a minimum sensor density sufficient for meaningful anomaly detection.
- Sensor sites have reasonably stable network or cellular connectivity for telemetry transmission.
- Each utility organization has at least one designated Administrator responsible for zone and user configuration.
- Third-party authentication, GIS, and storage providers maintain their own agreed service-level agreements (SLAs).

13. Constraints

Type	Constraint
Technical	Leak-detection accuracy depends on sensor density and the availability of sufficient baseline historical data per zone.
Operational	The system must remain available 24/7 since leaks and pressure events can occur at any time, across multiple shifts.
Economic	Cloud compute, storage, and sensor connectivity costs scale with the number of sensors and zones, requiring a data-retention policy.
Legal	The system must comply with data-protection regulations and any applicable critical-infrastructure security requirements.
Time-Related	The initial release must prioritize core monitoring, ingestion, and leak-detection features within the project timeline before advanced predictive-maintenance analytics.

14. Requirement Validation — Completeness and Consistency

14.1 Stakeholder Coverage Check

Each identified stakeholder (Control Room Operator, Field Technician, Utility/Asset Manager, GIS Engineer, Administrator, Regulatory Authority/Guest) is mapped to at least one functional requirement and one use case, confirming that no stakeholder's core need has been left unaddressed.

14.2 Ambiguity, Redundancy and Inconsistency Review

- Ambiguity: Terms such as 'statistical anomaly' (FR-06) and 'peak load' (Scalability NFR) are flagged as needing quantified thresholds in the next design iteration (e.g., defining the anomaly threshold as a specific standard-deviation bound on minimum night flow).
- Redundancy: FR-04 (Real-Time Monitoring Dashboard) and FR-14 (GIS Network Visualization) were reviewed to confirm they serve distinct purposes (live metric tiles vs. spatial pipeline context) rather than duplicating functionality.
- Inconsistency: Role terminology was standardized across FR-17, the Actors table, and the Non-Functional Security requirement to consistently use the roles Admin, Operator, Field Technician, Manager, and Guest.
- Missing Requirements Identified During Review: an explicit sensor-health/fault-flagging requirement (added as FR-05) and an audit-logging requirement (added under Data Requirements and the Auditability NFR) were included after the initial pass.

14.3 Suggested Improvements

- Quantify vague terms (e.g., define exact anomaly-detection thresholds and sensor-offline timeout values) in the detailed design phase.
- Add explicit acceptance criteria for each functional requirement to support test-case generation.
- Introduce a formal requirement traceability matrix linking each FR/NFR to its corresponding use case and test case.
- Conduct a stakeholder walkthrough/review session with utility operators and field crews before moving to the design phase to validate assumptions.

15. Conclusion

This scenario-based requirement analysis has identified the core functional and non-functional needs of an Intelligent Water Distribution Monitoring and Leakage Detection Platform, covering real-time sensor monitoring, automated leak detection and localization, predictive maintenance, GIS visualization, and alert/work-order management. The identified actors, use cases, and data requirements provide a solid foundation for the system's design phase, while the documented assumptions, constraints, and validation review help ensure the specification is complete, consistent, and ready for stakeholder sign-off before development begins.